

1. User pain

Students read lecture PDFs in VLearn but ask short or vague questions.

A generic chatbot may answer without knowing the current slide.

Risk: students trust unsupported answers and study the wrong thing.

2. Chosen slice

One student, one PDF page, one tutor decision.

The tutor answers from current slide context, asks for clarification, or refuses unsupported / restricted requests.

3. Prototype

Left: material list by day.

Middle: PDF slide viewer and master navigation bar.

Right: contextual tutor using DeepSeek when API key is configured.

4. Eval result

Golden set: 67 cases.

First baseline: 52/67.

Quality bar: at least 80%, and zero unsupported hallucination.

5. Validation

Run five user-test sessions.

Observe navigation, context copy/selection, and refusal behavior.

Log every issue in `validation/feedback_log.md`.

6. One more week

Add OCR for image-only slides.

Add automated eval runner and richer citations.

Persist notes/highlights beyond one Streamlit session.